

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0774

Biddeford Ip 100 kV
October 02, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0774. The Interconnection Customer proposes a 213.6 MW solar photovoltaic facility interconnecting at the Biddeford Ip 100 kV substation in the Illinois area. The request was received on April 11, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 303 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,122,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Walden / Invenergy Nelson Expansion Llc 100.0 kV	Taylorville S 230 kV/1 kV Transformer	237.7	137	99.9
Invenergy Nelson Expansion Llc / Ault 100.0 kV	Taylorville S 230/1 kV TX	233.0	146	99.9
Taylorville S 100 kV/1 kV XFMR	Ault / Invenergy Nelson Expansion Llc 100.0 kV	215.2	158	99.9
Biddeford Ip / Fairbanks 100.0 kV	Taylorville S 230 kV/1 kV Auto	206.0	146	100.0
Taylorville South 230 kV/1 kV Auto	Ault / Invenergy Nelson Expansion Llc 100.0 kV line	185.8	183	99.9
Walden / W Amesbury 100.0 kV Line	Taylorville S 230 kV/1 kV Autotransformer	181.6	146	71.5
Taylorville South / Biddeford Ip 100.0 kV	Invenergy Nelson Expansion Llc / Ault 100.0 kV	173.0	146	99.9
Alder / Shady Oaks Wind Farm 230.0 kV	Shady Oaks Wind Farm / Alder 500.0 kV	102.7	856	14.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Invenenergy Nelson Epansion Llc / Waldn 100.0 kV	Rebuild line with higher-capacity conductor	\$2,991,000	70% (\$2,092,000)
Invenenergy Nelson Expansion Llc / Ault 100.0 kV	Rebuild line with higher-capacity conductor	\$6,346,000	70% (\$4,437,000)
Taylorville S 100 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$3,034,000	70% (\$2,121,000)
Fairbanks / Biddeford Ip 100.0 kV	Rebuild line with higher-capacity conductor	\$10,655,000	70% (\$7,450,000)
Taylorville South 230-1 kV XFMR	Replace transformer with higher-rated unit	\$6,191,000	70% (\$4,329,000)
West Amesbury / Walden 100.0 kV L	Rebuild line with higher-capacity conductor	\$4,621,000	70% (\$3,231,000)
Biddeford Ip / Taylorville S 100.0 kV	Rebuild line with higher-capacity conductor	\$2,134,000	100% (\$2,134,000)
Shady Oaks Wind Farm / Alder 230.0 kV	Reconductor line	\$6,124,000	54% (\$3,328,000)
	Total allocated to project		\$29,122,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.